

Supplementary materials

Potential catecholaminergic pathway of volatility estimation in attention-deficit/hyperactivity disorder

Plank et al.

2026-08-25

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S1 Package versions

The following packages are used in this RMarkdown file:

```
## [1] "R version 4.5.2 (2025-10-31)"
## [1] "knitr version 1.51"
## [1] "ggplot2 version 4.0.3"
## [1] "brms version 2.23.0"
## [1] "designr version 0.1.13"
## [1] "bridgesampling version 1.2.1"
## [1] "tidyverse version 2.0.0"
## [1] "ggpubr version 1.0.0"
## [1] "gggrain version 0.1.2"
## [1] "bayesplot version 1.15.0"
## [1] "SBC version 0.5.0.9000"
## [1] "rstatix version 1.1.0"
## [1] "BayesFactor version 0.9.12.4.8"
## [1] "easystats version 0.7.6"
```

S2 Methods: additional details

S2.1 Procedure

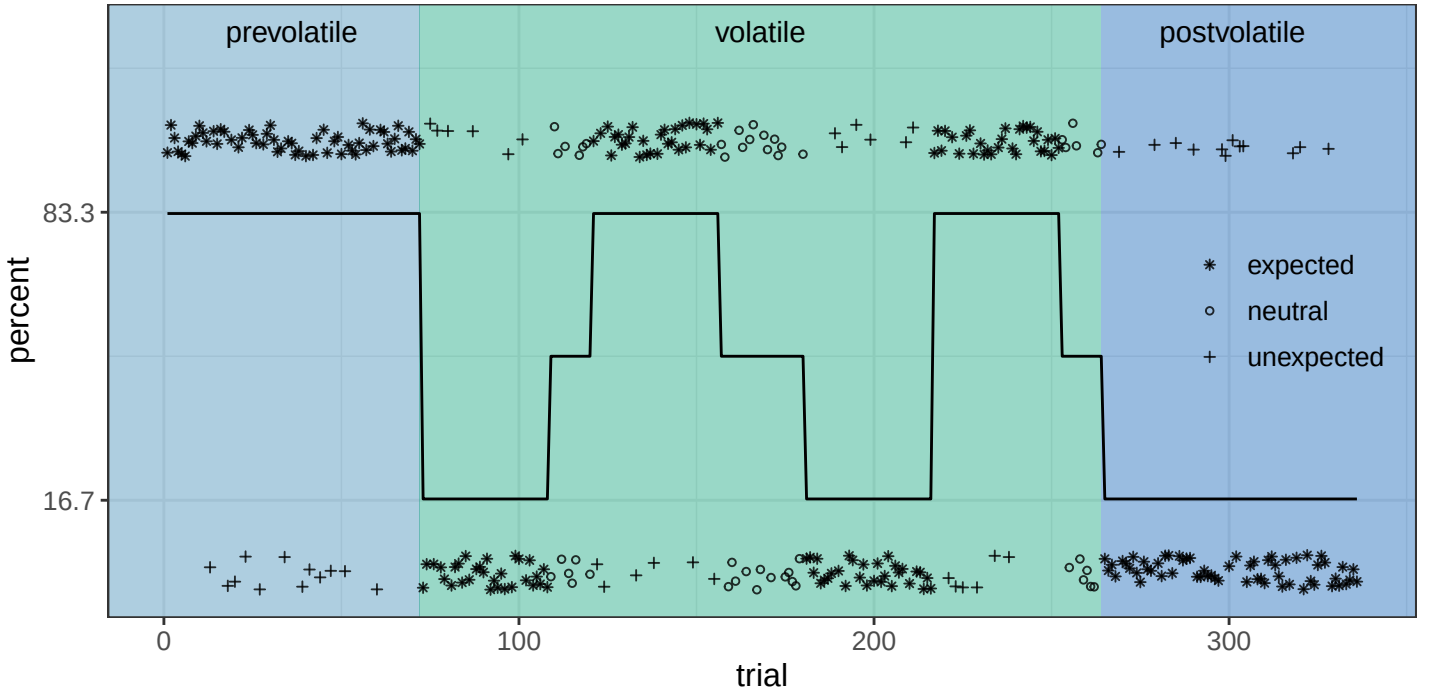
Behavioural tasks were performed in a small, soundproof testing chamber with consistent lighting on a Lenovo Legion Pro 5 mounted in a laptop stand (resolution: 2560x1600 pixel; size: 344x215mm; refresh rate: 60Hz). Eye-tracking data was collected with a LiveTrack Lightning eye tracker (Cambridge RS, sampling rate 500Hz) using a headrest to ensure a stable viewing distance of 57cm. The eye tracker was individually calibrated before each task based on 9-points.

S2.2 Stimulus creation

Stimuli were created using KDEF pictures of people portraying three expressions: emotionally neutral, fearful and happy (Lundqvist et al., 1998). Happy and fearful pictures were morphed with emotionally neutral pictures of the same person using WinMorph 3.01 to create videos of the person gradually and continually taking on an emotional expression. Independent samples of 21 and 25 participants rated the naturalness of videos of female-presenting and male-presenting faces, respectively (online surveys on SoSci, Leiner, 2019). We selected six identities based on these ratings (F02, F05, F09, M10, M11 and M13; naturalness of selected videos: range = 70.04-82.81; mean = 76.55). Of these videos, we extracted still images capturing 50%, 75% and 100% of the emotional expression to create hard, medium and easy stimuli, respectively.

S2.3 Development of associations across PAL task

High tone > positive, low tone > negative emotion



S2.4 Computational modelling of reaction times: Hierarchical Gaussian Filter

To ensure model reliability and faithfulness, we followed the Bayesian workflow adaptation for computational modelling as proposed by Hess and colleagues (2025). This workflow includes the computation of empirical priors, in our case based on the publicly available data from Lawson et al. (2021) as well as our own pilot data, simulation-based evaluation of parameter recovery and model identifiability, posterior predictive checks and model comparison.

S2.5 Comparison of LME across groups

```
##                bf          error          time          code
## diagnosis -0.5840522 9.993788e-05 Thu Oct 30 11:19:46 2025 eca1873f721
```

S2.6 Analysis of extracted HGF parameters

We analysed HGF parameters using a Bayesian linear model with a Gaussian distribution and included group as the population-level predictor. Additionally, we explored whether ADHD diagnosis or ADHD medication could be predicted

using scaled HGF parameters by fitting two Bernoulli models. For these Bernoulli models, we did not distinguish between adults with ADHD alone and those with comorbid ASD. We used a lognormal Bayesian linear mixed model to assess the influence of alpha level (second, third level), change (prevolatile to volatile, volatile to postvolatile) and their interaction on learning rate updates. The model also includes participants (level and change slopes) on the group-level.

S2.7 Computational modelling of reaction times: Wiener Diffusion Model

We used the ggdmc package to implement a Bayesian version of a hierarchical Wiener diffusion model (Lin & Strickland, 2020), since Bayesian versions of DDM are more robust to designs with fewer trials (Myers et al., 2022). We constructed three separate hierarchical DDMs, one for each group, as a single model for all groups did not converge. Each model included one drift rate per task phase (prevolatile, volatile, postvolatile) to assess differences between task phases with high and low environmental volatility. We assessed model reliability with simulation-based parameter recovery. Only models where all potential scale reduction factors assessing convergence of the Markov chains were smaller than 1.1 were included in the analysis (Gelman et al., 2003). To achieve convergence, we increased the thinning until stable models were obtained. Then, we extracted participant-specific DDM parameters and analysed the drift rates with a Bayesian linear mixed model with a Gaussian distribution (population-level predictors group, phase and their interaction; group-level intercept for participants). A similar model focusing on adults with ADHD, regardless of comorbid ASD, investigated the influence of medication on drift rates.

S2.8 Conventional analysis of reaction time variance, reaction times and error rates

We included only reaction times on correct trials in the analysis of the reaction times and reaction time variances. Additionally, we excluded extreme reaction times defined by the interquartile method. Participants whose accuracy was below 66.67% were excluded from all analyses. Additionally, reaction time variance was only computed as standard deviation for conditions where at least two-thirds of the trials' reaction times were included in the analysis. We used a shifted lognormal distribution to model correct reaction times with the model including group, expectancy, phase and difficulty on the population level. On the group level, we added trials (slope for group) and participants (slopes for expectancy, phase, difficulty and their interactions). We analysed reaction time variance with a lognormal model including the population-level predictors group, expectancy, phase and their interaction as well as participants with slopes for expectancy and phase on the group-level. We dropped the predictor difficulty due to poor posterior predictive fit, slightly deviating from the preregistration. Additionally, we used a Bayesian ANOVA on the rank transformed accuracies to explore differences.

S3 Participant-specific HGF and DDM parameters

S3.1 H3c: second level tonic volatility

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	-5.93	0.95	-6.48	-5.37	1.00	1	6174
b_diagnosis1	conditional	0.19	0.95	-0.39	0.78	0.74	1	6186
b_diagnosis2	conditional	0.15	0.95	-0.45	0.73	0.69	1	5704
sigma	sigma	2.28	0.95	1.99	2.61	1.00	1	5792

S3.2 Predicting ADHD diagnosis with HGF parameters

Parameter	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	0.06	0.95	-0.52	0.63	0.57	1	5569
b_sbe1	0.18	0.95	-0.53	0.91	0.70	1	5518
b_sbe2	-0.10	0.95	-0.81	0.65	0.60	1	6430
b_sbe3	-0.18	0.95	-0.89	0.51	0.69	1	6120
b_sze	0.31	0.95	-0.48	1.17	0.78	1	5970
b_som2	0.65	0.95	-0.01	1.40	0.97	1	6381
b_som3	-0.03	0.95	-0.89	0.82	0.53	1	6332

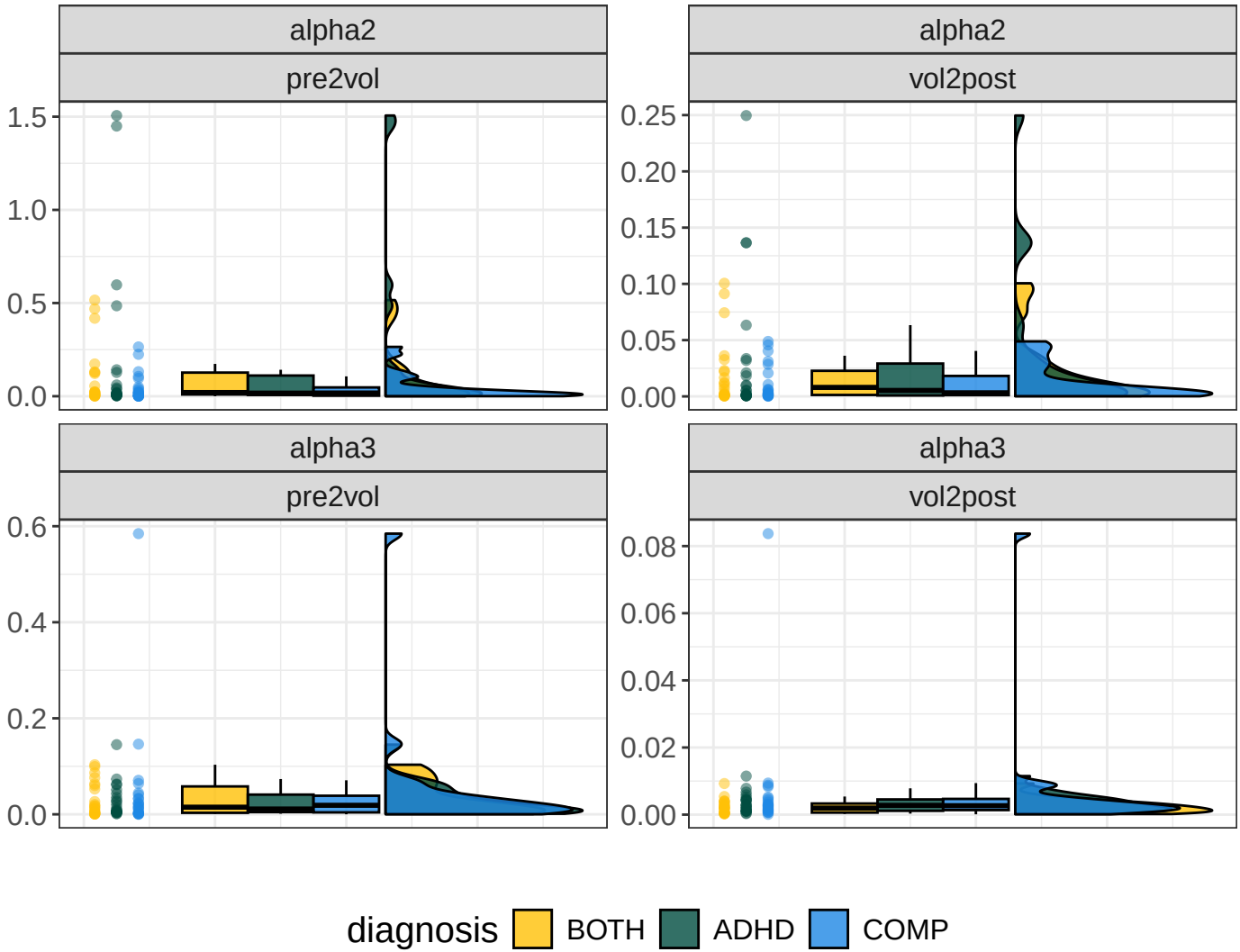
S3.3 Predicting ADHD medication with HGF parameters

Parameter	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	0.39	0.95	-0.17	0.96	0.91	1	5726
b_sbe1	0.18	0.95	-0.49	0.90	0.70	1	6722
b_sbe2	0.27	0.95	-0.43	1.00	0.77	1	6075
b_sbe3	0.04	0.95	-0.60	0.69	0.55	1	6472
b_sze	0.20	0.95	-0.46	0.91	0.72	1	6181
b_som2	-0.51	0.95	-1.20	0.11	0.95	1	6430
b_som3	-0.18	0.95	-0.94	0.56	0.69	1	5957

S3.4 Learning rate update

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	-4.95	0.95	-5.24	-4.66	1.00	1	2760
b_diagnosis1	conditional	0.18	0.95	-0.22	0.59	0.81	1	2534
b_diagnosis2	conditional	-0.06	0.95	-0.48	0.33	0.62	1	2841
b_level1	conditional	0.33	0.95	0.10	0.56	1.00	1	3706
b_change1	conditional	0.77	0.95	0.64	0.90	1.00	1	5573
b_diagnosis1:level1	conditional	0.06	0.95	-0.26	0.39	0.65	1	4210
b_diagnosis2:level1	conditional	0.15	0.95	-0.17	0.47	0.83	1	4199
b_diagnosis1:change1	conditional	0.05	0.95	-0.14	0.24	0.69	1	5829
b_diagnosis2:change1	conditional	0.04	0.95	-0.15	0.23	0.66	1	5894
b_level1:change1	conditional	-0.10	0.95	-0.22	0.03	0.94	1	5555
b_diagnosis1:level1:change1	conditional	0.06	0.95	-0.11	0.24	0.77	1	6185
b_diagnosis2:level1:change1	conditional	-0.07	0.95	-0.24	0.10	0.79	1	5620
sigma	sigma	1.01	0.95	0.89	1.15	1.00	1	4036

Learning rate updates



S3.5 Group differences in drift rate

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	2.31	0.95	2.16	2.47	1.00	1.01	1772
b_diagnosis1	conditional	-0.27	0.95	-0.48	-0.07	0.99	1.01	1820
b_diagnosis2	conditional	-0.05	0.95	-0.26	0.16	0.67	1.01	1974
b_phase1	conditional	-0.03	0.95	-0.09	0.04	0.78	1.00	5825
b_phase2	conditional	0.04	0.95	-0.02	0.11	0.90	1.00	5664
b_diagnosis1:phase1	conditional	0.08	0.95	-0.01	0.17	0.95	1.00	5891
b_diagnosis2:phase1	conditional	-0.01	0.95	-0.11	0.08	0.60	1.00	5845
b_diagnosis1:phase2	conditional	0.01	0.95	-0.08	0.11	0.60	1.00	5318
b_diagnosis2:phase2	conditional	-0.03	0.95	-0.12	0.06	0.74	1.00	5486
sigma	sigma	0.33	0.95	0.29	0.38	1.00	1.00	5283

S3.6 Influence of medication on drift rate

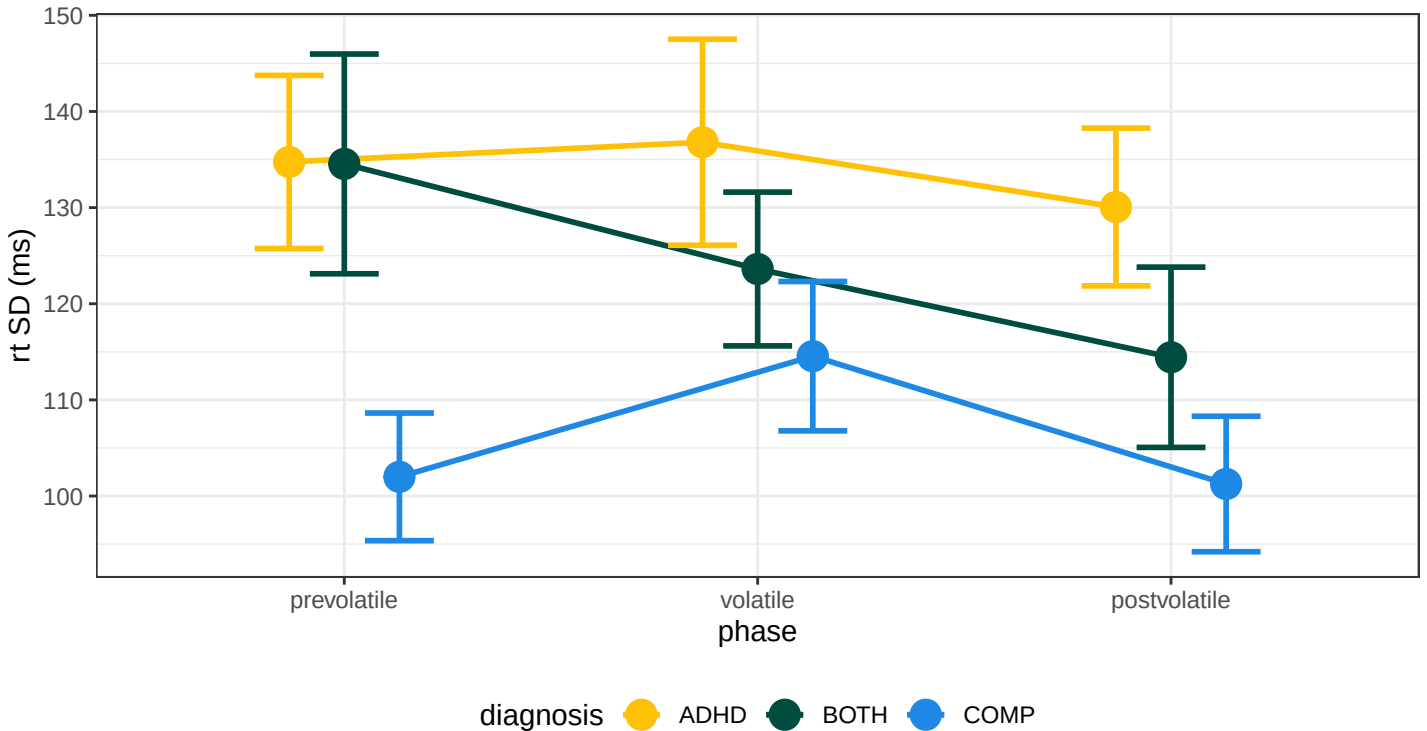
Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	2.14	0.95	1.97	2.32	1.00	1	2335
b_adhd.meds.bin1	conditional	0.06	0.95	-0.11	0.22	0.75	1	1984
b_phase1	conditional	0.01	0.95	-0.06	0.07	0.57	1	6272
b_phase2	conditional	0.03	0.95	-0.03	0.10	0.84	1	6017

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_adhd.meds.bin1:phase1	conditional	0.01	0.95	-0.06	0.08	0.65	1	5843
b_adhd.meds.bin1:phase2	conditional	0.00	0.95	-0.07	0.06	0.55	1	5958
sigma	sigma	0.28	0.95	0.24	0.32	1.00	1	5476

S4 Conventional analyses of responses and pupil sizes

S4.1 Response time variance

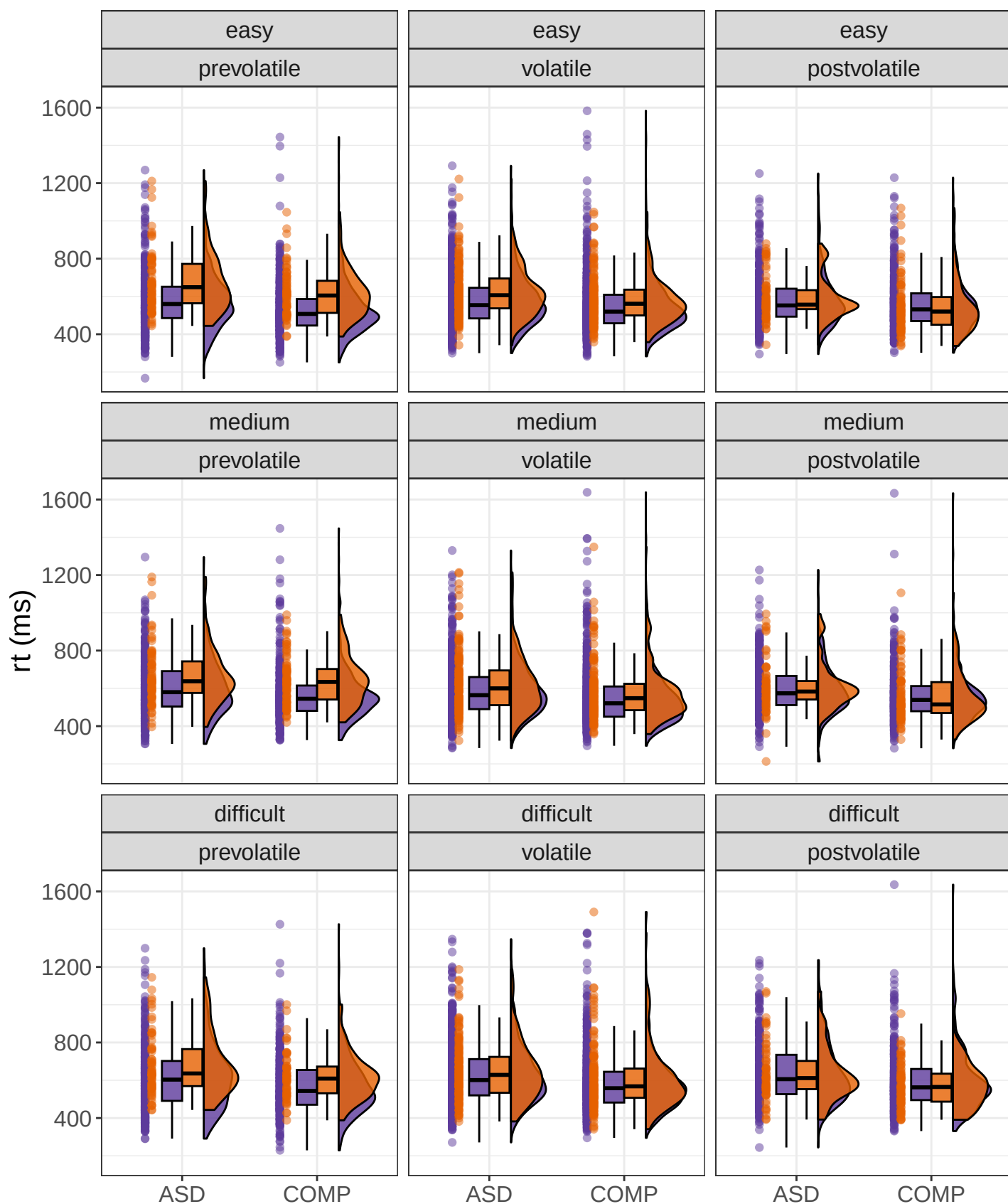
Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	4.71	0.95	4.62	4.79	1.00	1	5022
b_diagnosis1	conditional	0.09	0.95	-0.03	0.21	0.93	1	4509
b_diagnosis2	conditional	0.03	0.95	-0.09	0.15	0.69	1	4865
b_expected1	conditional	0.06	0.95	0.04	0.08	1.00	1	13918
b_phase1	conditional	0.02	0.95	-0.01	0.05	0.88	1	13959
b_phase2	conditional	0.03	0.95	0.00	0.06	0.98	1	13147
b_diagnosis1:expected1	conditional	-0.01	0.95	-0.04	0.02	0.75	1	14370
b_diagnosis2:expected1	conditional	0.00	0.95	-0.03	0.02	0.55	1	14145
b_diagnosis1:phase1	conditional	0.00	0.95	-0.05	0.05	0.50	1	13651
b_diagnosis2:phase1	conditional	0.05	0.95	0.00	0.10	0.97	1	13448
b_diagnosis1:phase2	conditional	-0.03	0.95	-0.07	0.01	0.94	1	13824
b_diagnosis2:phase2	conditional	-0.02	0.95	-0.06	0.02	0.84	1	14268
b_expected1:phase1	conditional	0.02	0.95	0.00	0.05	0.96	1	15127
b_expected1:phase2	conditional	-0.02	0.95	-0.05	0.00	0.96	1	15707
b_diagnosis1:expected1:phase1	conditional	-0.01	0.95	-0.05	0.03	0.68	1	15059
b_diagnosis2:expected1:phase1	conditional	-0.01	0.95	-0.05	0.02	0.77	1	13226
b_diagnosis1:expected1:phase2	conditional	0.04	0.95	0.00	0.07	0.97	1	14941
b_diagnosis2:expected1:phase2	conditional	0.00	0.95	-0.03	0.04	0.60	1	14865
sigma	sigma	0.18	0.95	0.17	0.20	1.00	1	10033



S4.2 Response times

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	6.17	0.95	6.12	6.23	1.00	1	4403
b_diagnosis1	conditional	0.03	0.95	-0.01	0.07	0.93	1	3305
b_expected1	conditional	-0.04	0.95	-0.05	-0.02	1.00	1	12702
b_phase1	conditional	0.03	0.95	0.00	0.05	0.99	1	9589
b_phase2	conditional	-0.01	0.95	-0.02	0.01	0.82	1	11148
b_difficulty1	conditional	-0.03	0.95	-0.04	-0.01	1.00	1	11626
b_difficulty2	conditional	-0.01	0.95	-0.02	0.01	0.86	1	11731
b_diagnosis1:expected1	conditional	0.00	0.95	-0.01	0.01	0.63	1	14759
b_diagnosis1:phase1	conditional	0.00	0.95	-0.02	0.02	0.50	1	10799
b_diagnosis1:phase2	conditional	0.00	0.95	-0.01	0.01	0.55	1	14633
b_expected1:phase1	conditional	-0.04	0.95	-0.05	-0.02	1.00	1	11343
b_expected1:phase2	conditional	0.00	0.95	-0.01	0.02	0.57	1	12070
b_diagnosis1:difficulty1	conditional	0.00	0.95	-0.01	0.01	0.56	1	14624
b_diagnosis1:difficulty2	conditional	0.00	0.95	-0.01	0.01	0.81	1	15227
b_expected1:difficulty1	conditional	-0.01	0.95	-0.03	0.00	0.97	1	11088
b_expected1:difficulty2	conditional	0.00	0.95	-0.01	0.02	0.59	1	10283
b_phase1:difficulty1	conditional	0.00	0.95	-0.02	0.02	0.58	1	11648
b_phase2:difficulty1	conditional	0.01	0.95	-0.01	0.03	0.78	1	11209
b_phase1:difficulty2	conditional	0.02	0.95	0.00	0.04	0.94	1	11425
b_phase2:difficulty2	conditional	-0.01	0.95	-0.03	0.01	0.87	1	11218
b_diagnosis1:expected1:phase1	conditional	0.01	0.95	0.00	0.02	0.85	1	15621
b_diagnosis1:expected1:phase2	conditional	0.00	0.95	-0.01	0.01	0.52	1	15510
b_diagnosis1:expected1:difficulty1	conditional	0.00	0.95	-0.01	0.01	0.62	1	14751
b_diagnosis1:expected1:difficulty2	conditional	0.00	0.95	-0.01	0.01	0.70	1	14948
b_diagnosis1:phase1:difficulty1	conditional	0.01	0.95	-0.01	0.02	0.82	1	14451
b_diagnosis1:phase2:difficulty1	conditional	0.00	0.95	-0.01	0.01	0.67	1	12915
b_diagnosis1:phase1:difficulty2	conditional	-0.01	0.95	-0.02	0.01	0.84	1	14790
b_diagnosis1:phase2:difficulty2	conditional	0.01	0.95	-0.01	0.02	0.85	1	14708
b_expected1:phase1:difficulty1	conditional	-0.01	0.95	-0.03	0.01	0.75	1	11261
b_expected1:phase2:difficulty1	conditional	0.00	0.95	-0.02	0.02	0.51	1	11295
b_expected1:phase1:difficulty2	conditional	0.01	0.95	-0.02	0.03	0.74	1	12268
b_expected1:phase2:difficulty2	conditional	0.00	0.95	-0.02	0.02	0.64	1	10958
b_diagnosis1:expected1:phase1:difficulty1	conditional	0.00	0.95	-0.01	0.02	0.69	1	13584
b_diagnosis1:expected1:phase2:difficulty1	conditional	0.00	0.95	-0.01	0.01	0.57	1	14873
b_diagnosis1:expected1:phase1:difficulty2	conditional	0.00	0.95	-0.01	0.01	0.53	1	14760
b_diagnosis1:expected1:phase2:difficulty2	conditional	0.00	0.95	-0.02	0.01	0.80	1	14645
sigma	sigma	0.23	0.95	0.22	0.24	1.00	1	13192

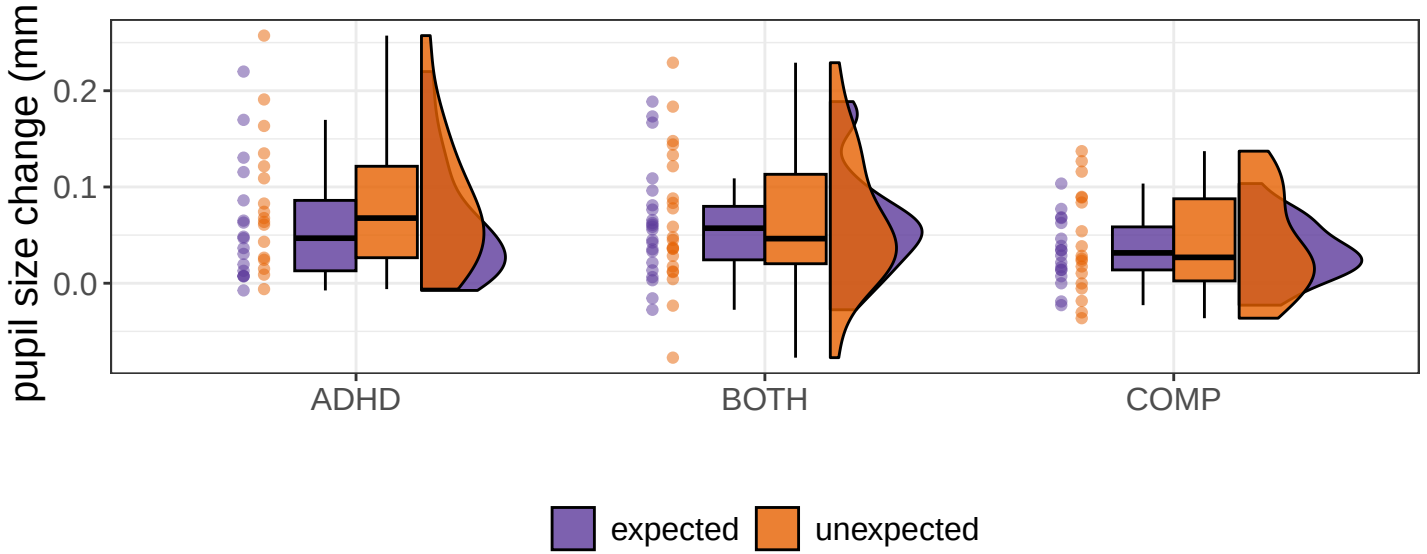
Reaction times per subject



expected unexpected

S4.3 Pupil sizes

Parameter	Component	Median	CI	CI_low	CI_high	pd	Rhat	ESS_tail
b_Intercept	conditional	0.06	0.95	0.04	0.07	1.00	1.01	2673
b_diagnosis1	conditional	0.01	0.95	-0.01	0.03	0.81	1.00	2062
b_diagnosis2	conditional	0.00	0.95	-0.02	0.03	0.64	1.00	1993
b_expected1	conditional	0.00	0.95	-0.01	0.00	0.91	1.00	12036
b_rts	conditional	0.00	0.95	0.00	0.00	1.00	1.00	15436
b_diagnosis1:expected1	conditional	0.00	0.95	-0.01	0.00	0.94	1.00	14814
b_diagnosis2:expected1	conditional	0.00	0.95	0.00	0.01	0.87	1.00	13960
sigma	sigma	0.18	0.95	0.17	0.18	1.00	1.00	11972



S4.4 Accuracies

Since the number of models which are compared is very high with 166, we only print the best 10 models.

```
## ---
## Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis : fixed
```

	bf
diagnosis + expected + difficulty	46.41
diagnosis + difficulty	46.25
diagnosis + expected + difficulty + diagnosis:difficulty	45.02
diagnosis + difficulty + diagnosis:difficulty	44.85
diagnosis + phase + expected + phase:expected + difficulty	44.77
expected + difficulty	44.42
difficulty	44.28
diagnosis + expected + difficulty + expected:difficulty	43.35
diagnosis + phase + expected + difficulty	43.23
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty	43.19

Accuracies per subject

